

Pulse: Hackathon Winning Plan

Inside the Game: Developer Hackathon | Team brainstorm edition (draft v2, Oct 2, 2026)

How to use this document

Read it end to end, then use the yellow "Discuss as a team" boxes to react, challenge and add ideas. Section 16 is the decision log we fill in together before work starts. Nothing here is final: the schemas, weights and scope are all open to change.

1. Where we stand

A team of three, building on Azure, entering the Microsoft x Premier League hackathon. The challenge is to turn synthetic football events into explainable match intelligence, real-time narratives and automated recaps, personalized for every fan and usable by studios and streaming platforms.

Key dates (EAT)

Date	Milestone
Sept 29	Registration opened
Oct 6, 7:00 PM	Submission period opens (9:00 AM PT)
Oct 20, 10:00 PM	Registration closes (12:00 PM PT). Register well before this.
Oct 22	Our own feature freeze
Oct 25	Our own target to submit
Oct 28, about 10:00 AM	Submissions close (Oct 27, 11:59 PM PT)
Nov 10	Judging ends. The demo must stay live and free for judges until then.
By Nov 20	Winners announced

Decisions already made

- Concept: **Pulse**, a match intelligence layer built around a Control vs Chaos index, serving both broadcast/overlay and fan experience.
- **Software only and web-based.** No ElevenLabs, no headset or glasses hardware, no broadcast hardware. Voice comes from Azure AI Speech.
- **Facts first, LLM second.** Stats and moments are computed deterministically. Agents narrate only from a structured fact pack, and a Verifier checks every claim.
- **Prize target:** one Grand Prize plus the Best Multi-Agent System category, with Azure Cloud Native as a natural second fit.

2. What the official rules mean for us

Rule	What we do about it
Microsoft hero tech must be central (Stage 1 pass/fail, plus 20% of the score)	Foundry, Microsoft Agent Framework and an MCP server at the core. Azure OpenAI through Foundry, Container Apps, Event Hubs, a managed database. Foundry IQ and Fabric are optional extras.
New projects only (created after Sept 29)	Fresh repository now. No code imported from earlier projects.
Public GitHub repo, and code quality is scored	No secrets in the repo (use Key Vault). Strong README, tests, linting, CI and an architecture doc.
Demo video under 2 minutes, public URL, no third-party trademarks or copyrighted music	Fictional clubs, players and crests only. No real club or player names, no Premier League marks, no music unless properly licensed.
The project must function as depicted in the video	Only show what we have really built. Report measured latency, not aspirational numbers.
Free, unrestricted access for judges until Nov 10	Scale-to-zero hosting, a replay-driven demo match, cached model output, spend alerts.
Hardware clause: judges may request physical access to wearable or proprietary hardware	Avoided entirely: the project is a web app.
All materials in English	English README, pitch and subtitles for any non-English clip.
Content restrictions (no alcohol, betting-style promotion or political agenda)	Keep the demo clean. No odds, no sponsor-style branding.
One Grand Prize and one Category Prize per entrant at most	Aim for Grand plus Multi-Agent. Judges may reassign categories, so keep the Azure story strong too.

Judging criteria (20% each): technological implementation (including creativity in synthetic data), agentic design and innovation, real-world impact, user experience and presentation (balanced frontend and backend), and adherence to the category.

3. The core idea: Pulse

Never let a model invent football. Compute everything deterministically, then let agents narrate only from a structured fact pack. Every sentence in a recap or overlay traces back to the events that justified it, which gives us explainability and a clear answer to hallucination.

The spine of the product is the **Control vs Chaos index**. Every insight, overlay and line of commentary is anchored to a swing in it, so “why this matters” has one consistent explanation across the whole product.

Tagline options

- Beyond the scoreline.

- Every moment, explained.
- The story behind every pass.
- Football, explained for every fan.

Discuss as a team

- Which tagline fits best, and do we want a product name other than Pulse?
- Does the Control vs Chaos framing feel strong enough as the headline of the demo?

4. Football intelligence: the metrics

These come from established football analytics ideas (pitch control in the style of Spearman, spatial influence work by Fernández and Bornn). We use simplified versions that run on a coarse grid every 500 ms, so they stay fast and explainable.

Metric	What it tells the viewer	How we compute it (version 1)
Pitch control	Who owns which space right now	For each grid cell (about 21 by 14), estimate which team reaches it first from player positions, speeds and reaction time, with a logistic uncertainty term. Result is 0 to 1 per cell.
Pressing intensity	How quickly the ball carrier is running out of time	Probability that at least one defender arrives within a 1.5 second horizon. Above about 0.75 is flagged as severe pressure (threshold is tunable).
Pass Difficulty Rating	How hard a pass really was	A logistic model over pass distance, pressure on the passer, nearest defender intercept time and lane clearance. Coefficients are tuned on our synthetic data and documented as heuristics.
Tactical entropy	Structure versus chaos	Average binary entropy across the pitch control grid, normalised to 0 to 1 . Low means stable shape, a spike means transition chaos.
Speed and distance flags	Sprints, running load, shot speed	Speed from position changes. Sprint flag above about 7 m/s (roughly 25 km/h) for two seconds or more. Ball and shot speed from ball position deltas.
Control vs Chaos index	The headline number for the whole product	A weighted blend of pitch control share, inverse entropy, pressing intensity and shot quality. Weights are tuned until the index tells the intended story on scripted storylines.

Discuss as a team

- Are we comfortable implementing these, and who owns the maths?
- How do we convince ourselves the index “feels right”? Proposal: validate it against the scripted storylines.

5. Synthetic data innovation

This is a judged criterion (creativity and optimised synthetic data creation) and also the foundation of the demo. The generator is **event-first**: scripted storylines produce possession chains and events, and a lightweight tracking layer (around 5 Hz, players moving around formation anchors with noise and ball attraction) supplies the positions needed for the metrics above. We deliberately skip a full physics simulation.

- **Storyline archetypes**: late siege, red-card swing, pressing trap, backs-to-the-wall defence, comeback. Seeded, so any match is reproducible for judges.
- **Realism checks**: distribution checks on pass accuracy, shots per match, possession ranges and sprint counts, published as a short data-quality report.
- **Published artefacts**: schema, generator CLI and sample matches in the public repo.
- **Fictional league world**: invented clubs, players, kits and venues, which also keeps us clear of trademark and likeness issues.

Draft telemetry fields (final Event and Cue schemas to be locked later)

Field	Type	Purpose
timestamp_utc	ISO-8601	Base clock for synchronisation
smpte_timecode	String	Frame-accurate alignment; used by the roadmap broadcast adapter
phase_id	ID	Groups events in one uninterrupted possession chain
event_type	Enum	Pass, Shot, Tackle, Duel, Carry, Pressure, and others
ball_coordinates	x, y, z	Pitch location (105 by 68 m) and height
ball_speed_mps	Float	Pass pace and shot speed
actor_id, recipient_id	Player IDs	Who did it, and who received it (if a pass)
pass_distance_m, pass_difficulty_pdr	Float	Distance and Pass Difficulty Rating
pressure_index	0 to 1	Pressing intensity on the ball carrier
team_pitch_control	0 to 1	Territorial control of the possessing team
tactical_entropy	0 to 1	Chaos versus structure (normalised)

Discuss as a team

- Which three storylines make the best demo match?
- What fictional league, clubs and players do we invent? Add names and a one-line identity for each club.

6. Multi-agent design

Agentic design is 20% of the score and Multi-Agent is our target category. The pipeline becomes a team of specialised agents on Microsoft Agent Framework, running on Foundry, coordinating through shared match state and an MCP server that exposes match facts and calculations as tools.

Agent	Role	Hands off
Orchestrator	Routes moments by priority and latency tier, owns shared state, handles retries and timeouts.	Assigns work to the agents below
Analyst	Reads match state through MCP tools and flags moments: momentum swings, milestones, speed and distance thresholds, pressing traps.	Moment with event IDs
Explainer	Adds the why: control vs chaos breakdown, tactical pressure change, game rhythm.	Insight with reasons
Narrator	Turns the fact pack into commentary beats and milestones, optionally grounded in synthetic season history through Foundry IQ.	Draft narration
Persona and Localisation	Re-voices the same facts per viewer profile and locale, generated from facts and a football terminology lexicon rather than literally translated.	Audience-specific narration
Producer	Builds timed, machine-readable cues for overlays, with priority, duration and speech markup.	Cue JSON
Verifier	Checks every claim against source event IDs. Rejects and sends back for a rewrite on failure.	Approved narration or fallback

Model plan

- A small, fast model for live narration and persona rewrites.
- A stronger model for the Explainer on major moments and for recaps, where a few seconds of latency is fine. Evaluate Foundry Model Router for choosing between them.

Failure recovery (a judging talking point)

- If the Verifier rejects a draft twice, the system falls back to a deterministic template sentence, so a wrong claim never reaches the screen.
- If a model call is slow or unavailable, the deterministic insight still ships immediately and narration streams in afterwards.
- Every handoff is traced in Foundry, so the demo can show a rejected claim and the recovery live.

Discuss as a team

- Is seven agents the right number, or should some be merged to keep the build manageable?
- Do we include Foundry IQ grounding in version 1 or treat it as a stretch?

7. Pipeline, delivery and latency

Stage	What it does	Azure building blocks
Ingest	Go service consuming synthetic events and tracking snapshots, replayable at 1x or 10x	Event Hubs (standard tier), Container Apps
Interpret	Rolling windows compute the metrics in section 4 and update match state	Go aggregation, Cosmos DB or Redis
Explain	Moment detectors with attached reasons, plus the Explainer and Verifier agents	Foundry, Azure OpenAI
Render	Cue JSON pushed to overlays and a fan dashboard shown beside a simulated feed	Web PubSub or SignalR, Next.js
Personalize	Persona and language agents, plus spoken commentary and recaps	Foundry, Azure AI Speech, Cosmos DB for profiles

Latency: promise only what we can measure

- **Deterministic insights** (metrics, moments, cues) should appear within a fraction of a second. This is our real-time claim.
- **Model narration** streams in after, typically within a second or two. We will measure and show real numbers in the demo and write-up.
- **Recaps** are generated at full time and shown with measured generation time. We will not claim fixed numbers until we have tested them.

How overlays reach the screen

- **Web delivery (what we build)**: cue JSON over Web PubSub, rendered as an HTML5 overlay on top of a video player. Cues are also carried as timed metadata in the video player, so they stay frame-aligned with playback.
- **Broadcast adapter (roadmap, not demoed)**: cues carry an SMPTE timecode and a documented adapter interface, so a future gateway could feed broadcast graphics engines. We will describe this honestly as roadmap because we have no broadcast hardware.

Discuss as a team

- Is browser timed metadata the right way to demonstrate sync, or is there a simpler approach for the demo?
- How much of the roadmap story belongs in the pitch without overclaiming?

8. Outputs and personalization

Three output tiers

- **Live lower-thirds:** short tactical updates of under 15 words for immediate on-screen display.
- **Commentary prompts:** conversational analysis for studio commentators, or spoken by our Commentary Booth voices.
- **Recaps:** halftime and post-match summaries with turning points, key moments and traceable claims.

Viewing profiles

Aspect	Casual supporter	Tactical analyst	Player spotlight
Spatial overlays	Hidden in play, shown at key moments	Continuous pitch control and passing lanes	Influence area and movement trail for one player
Pass quality	Simple tags such as Sharp Pass or Risky Delivery	Difficulty rating and value change on every pass	Full pass map with difficulty colouring
Speed and physical	Badge when a sprint threshold is hit	Velocity, acceleration, high-intensity distance	Live sprint count and distance gauge
Tone	Conversational, dramatic, milestone-focused	Analytical and systemic	Individual execution and positional impact
Placement	Minimal lower-third	Multi-panel sidebar with pitch minimap	Retractable player panel over the video

- **Newcomer mode:** explains football concepts as they happen, for people new to the game.
- **Club Loyalist mode:** frames the same moment from the viewer's club, so a defensive stand reads as resilience for one side and as probing pressure for the other.
- **Languages:** English plus Swahili as the showcase, generated from facts with a football terminology lexicon. We only ship languages we can quality-check.

Discuss as a team

- Which profiles must be in version 1? Proposal: Casual, Analyst and Player Spotlight, plus Swahili.
- Who can review Swahili output, and is there a second language someone can review?

9. Creative ways to stand out (software only)

Idea	What the viewer sees	Why it scores	Effort
Evidence Trail	Click any insight to see "why am I seeing this?": a mini pitch replays the 5 to 10 events behind it and highlights the index swing.	Explainability, trust, proof for the Verifier	Medium

Idea	What the viewer sees	Why it scores	Effort
Commentary Booth	Two Azure AI Speech voices (play-by-play and analyst) with word-synced captions on the overlay.	UX, personalization, accessibility	Medium
Persona modes	Casual, Analyst, Newcomer and Club Loyalist views of the same moment.	Genuinely different experiences, as stage (e) asks	Medium
Swahili-first showcase	Commentary generated natively in Swahili, reviewed by a native speaker.	Multi-language, fans in any market, distinctive	Small to medium
Player Lock	Follow one player: involvement, pass difficulty, speed and distance alerts, a running player story.	Stage (e) and real-time intelligence	Medium
Match Radio	A 60 to 90 second personalised audio recap with captions.	Automated recaps, accessibility	Small
Storyline Generator	Seeded synthetic matches with archetypes, realism checks, published schema and CLI.	Creativity in synthetic data	Medium
Producer Rundown	A studio view: upcoming cues on a timeline, talking points for the next 60 seconds, approve or skip.	Broadcast angle, enterprise readiness	Medium
Heads-up mode	A glanceable mobile or second-screen layout showing one card at a time.	UX, no hardware needed	Small
Accessibility mode	Audio-first description, high contrast, screen-reader friendly captions, verbosity control.	Grand Prize wording mentions accessible experiences	Small
Responsible AI panel	Per-recap groundedness score, verified claims count, latency and cost.	Enterprise prize, Foundry evaluation story	Small to medium

Hero trio for the demo: Evidence Trail, Commentary Booth and Persona modes with the Swahili showcase. The **Storyline Generator** underpins all of it, because scripted storylines guarantee interesting moments on cue.

Parked for now: video auto-eventing (circular on synthetic data we generate ourselves), a full physics simulation, broadcast hardware integration (SMPTE gateways, graphics engines), native device apps, voice Q&A, and any language we cannot quality-check.

Discuss as a team

- What is the single most memorable thing a judge should see in the first 30 seconds?
- What idea is missing from this list that would make a judge say “nobody else did that”?

10. Scope: must, nice, park

Tier	Items
Must have	All five pipeline stages working live; metrics and Control vs Chaos index; the agent team with the Verifier loop; Evidence Trail; Casual, Analyst and Player Spotlight views; English and Swahili; spoken commentary with synced captions; storyline generator; deployment on Azure; public repo with tests and docs.
Nice to have	Producer Rundown; Match Radio; Heads-up mode; accessibility mode; Responsible AI panel; Newcomer and Club Loyalist modes; Foundry IQ grounding; Fabric in the data path.
Park	Video auto-eventing; full physics simulation; broadcast hardware adapters; device apps; voice Q&A; extra languages.

11. Engineering standards

Code quality is part of the score, so we set the standards on day one.

- **Domain model first:** entities, relationships and boundaries agreed before code. Schemas shared across services (for example Zod on the TypeScript side with matching Go types).
- **Languages:** Go for ingest and aggregation, TypeScript for the front end, shared tooling and agent glue where it fits.
- **Quality gates:** ESLint and Prettier for TypeScript, golangci-lint and gofmt for Go, automated tests, and CI on every pull request.
- **Tests that matter:** replay-based golden tests for the engine, plus an evaluation suite checking that every generated claim maps to a source event.
- **Infrastructure as code** and documentation: README, architecture diagram and short decision records.

Discuss as a team

- What branching model, review rules and repository layout do we use?
- Which licence for the public repo?

12. Team split

Role	Owns	Also
1. Engine and data	Go ingest and aggregation, storyline generator, metrics, control index and moment detectors, Azure infrastructure and CI	MCP server exposing match facts
2. Agents and AI	Agent Framework orchestration, Narrator, Persona and Localisation, Verifier, evaluations and groundedness checks	Speech integration and caching
3. Experience	Overlay renderer, fan dashboard, Evidence Trail, profile views, accessibility, demo video	README polish and architecture diagrams

Teams can have up to four members. A fourth person focused on design and the demo video would help, if we know someone eligible.

Discuss as a team

- Who takes which role, and where are the overlaps that need a backup?
- Who is our appointed Representative?

13. Timeline (dates in EAT)

Dates	Focus	Exit criteria
Oct 2 to 5	Register, appoint the Representative, create the fresh public repo, set up Azure and a Foundry project, draft schemas, start the generator.	Registered; repo live with CI; generator producing a basic match
Oct 6 to 12	First end-to-end slice: event, moment, cue, on screen. Metrics and control index working.	A synthetic match replays and a moment appears on screen
Oct 13 to 19	Agent team, profiles, Swahili, speech with captions, Evidence Trail, Player Lock.	Full demo path works from start to finish
Oct 20 to 23	Hardening: evaluations, cost controls, accessibility, deploy, judge access. Feature freeze Oct 22.	Public URL stable; tests green; spend alerts set
Oct 24 to 25	Record the video, write the pitch, final README, submit.	Submitted by Oct 25
Oct 26 to 27	Buffer only. Real deadline is Oct 28, about 10:00 AM EAT.	Nothing new, only fixes

14. Demo video beat sheet (under 2 minutes)

The match uses invented clubs and players only. No real club, player or league branding appears in the video.

1. **0:00 to 0:10:** the problem in one line: a scoreline hides the story. Introduce Pulse.
2. **0:10 to 0:30:** a synthetic match running at 10x, the Control vs Chaos index swinging, a moment firing on the overlay with a player speed tag.
3. **0:30 to 0:50:** Evidence Trail: tap the insight and see the events and reasons behind it.
4. **0:50 to 1:10:** the same moment for an analyst and a casual fan side by side, then in Swahili with spoken commentary and synced captions (English subtitles).
5. **1:10 to 1:25:** Player Spotlight and, if built, the Producer Rundown.
6. **1:25 to 1:45:** the agent trace: the Verifier rejects a bad claim and the system recovers. One frame of the Azure architecture.

7. **1:45 to 1:55:** the auto-generated recap, with a measured generation time, and a closing line on real-world impact.

15. Submission checklist and risks

Checklist

- Pitch naming what we built, the Microsoft and Azure technologies used, what it does and what problem it solves
- Demo video under 2 minutes with a public URL (YouTube, Vimeo, Facebook Video or Youku), showing the project running on its intended platform
- Public GitHub repository link
- A working link for judges, with login details if private, free to use until Nov 10
- Everything in English, with no third-party trademarks or copyrighted music in the video
- One appointed Representative who submits for the team, and no personal data of team members in the submission
- Nobody on the team employed by, or related to staff of, Microsoft or the Premier League (see the eligibility section of the rules)

Risks and mitigations

Risk	Mitigation
Scope creep across too many ideas	Hold to the hero trio. Everything else is nice-to-have. Feature freeze on Oct 22.
Hallucinated football facts	Fact pack plus the Verifier loop and template fallback; show groundedness in the demo.
Over-promising latency or broadcast capability	Claim only what we measure and demonstrate. Describe broadcast integration as roadmap.
Azure cost or quota during judging	Standard-tier services, scale-to-zero, caching, replay-driven demo, spend alerts, cheaper fallback model.
Swahili or other language quality	Native-speaker review on the team. Ship only languages we can quality-check.
Judges cannot reproduce the demo	Deterministic seeded matches, a one-click demo mode and clear testing instructions.
Real names or logos slipping into the video	Fictional league world from day one; checklist review before recording.
Metric formulas look impressive but do not feel right	Validate the index against scripted storylines; keep formulas simple and explainable.

16. Decision log and open questions

Fill this in together in the first team session. Add rows as needed.

Decision	Options / notes	Owner	Decide by
Product name and tagline	Pulse? Tagline options in section 3		Oct 3
Representative for the submission			Oct 3
Role assignments	See section 12		Oct 3
Fictional league: clubs, players, kits	Section 5		Oct 4
Three demo storylines	Late siege, red-card swing, pressing trap, comeback, other?		Oct 4
Control vs Chaos index weights	Start simple; validate on storylines		Oct 8
Languages beyond English and Swahili	Who can quality-check?		Oct 8
Foundry IQ and Fabric: in or out	Stretch or version 1?		Oct 8
Models for live narration versus recaps	Evaluate Model Router		Oct 10
Repo name, licence, branching and CI	Section 11		Oct 3
Add a fourth team member?	Design and video focus		Oct 5
Event and Cue schemas	Draft fields in section 5; lock when ready		When ready

Open brainstorm questions

1. What would make a judge remember our project a week later?
2. Where could the Verifier loop be made even more visible and convincing?
3. What is the most compelling personalization we have not listed yet?
4. What is the biggest technical risk, and what is our plan B?
5. If we had to cut half of the must-haves, which half goes?